

Arjuna JEE 2025

Maths

DPP: 3

Trigonometric Equations

Q1 If $\sin 2x = 4 \cos x$, then x is equal to -

- (A) $\frac{n\pi}{2} \pm \frac{\pi}{4}, n \in \mathbb{Z}$
 (B) no value
 (C) $n\pi + (-1)^n \frac{\pi}{4}, n \in \mathbb{Z}$
 (D) $2n\pi \pm \frac{\pi}{2}, n \in \mathbb{Z}$

Q2 The number of solution in $[0, 2\pi]$ of the equation $16^{\sin^2 x} + 16^{\cos^2 x} = 10$, is

- (A) 2 (B) 4
 (C) 6 (D) 8

Q3 Solve the equation & find the value of x
 $\tan\left(\frac{\pi}{4} - x\right) + \tan\left(\frac{\pi}{4} + x\right) = 4$

- (A) $x = n\pi \pm \frac{\pi}{3}$
 (B) $x = n\pi \pm \frac{\pi}{6}$
 (C) $x = n\pi + \frac{\pi}{3}$
 (D) $x = n\pi + \frac{\pi}{6}$

Q4 If $\sin 6\theta + \sin 4\theta + \sin 2\theta = 0$, Then the general value of θ is -

- (A) $\frac{n\pi}{4}, n\pi \pm \pi/3$
 (B) $\frac{n\pi}{4}, 2n\pi \pm \pi/3$
 (C) $\frac{n\pi}{4}, n\pi \pm \pi/6$
 (D) $\frac{n\pi}{4}, 2n\pi \pm \pi/6$

Q5 The solution set of the equation $4\sin^4 x + \cos^4 x = 1$ is

- (A) $x = n\pi + \frac{\pi}{6}$
 (B) $x = n\pi \pm \theta$ where $\cos^2 \theta = \frac{3}{5}$ and $x = n\pi, n \in \mathbb{I}$
 (C) $x = (2n+1)\frac{\pi}{2}$
 (D) $x = 2n\pi \pm \frac{\pi}{3}$

Q6

If $\sin \theta + \cos \theta = 1$ then the general value of θ is

- (A) $2n\pi$
 (B) $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{4}$
 (C) $2n\pi + \frac{\pi}{2}$
 (D) None of these

Q7 Find the general solution of $\cos 3\theta = \sin 2\theta$

- (A) $\theta = (4n+1)\frac{\pi}{10}$ and $\theta = (4n-1)\frac{\pi}{10}$
 (B) $\theta = (4n+1)\frac{\pi}{10}$ and $\theta = (4n-1)\frac{\pi}{2}$
 (C) $\theta = (4n+1)\frac{\pi}{2}$ and $\theta = (4n-1)\frac{\pi}{10}$
 (D) $\theta = (4n+1)\frac{\pi}{2}$ and $\theta = (4n-1)\frac{\pi}{2}$

Q8 A value of θ satisfying $\cos \theta + \sqrt{3} \sin \theta = 2$ is

- (A) $\frac{5\pi}{3}$
 (B) $\frac{4\pi}{3}$
 (C) $\frac{2\pi}{3}$
 (D) $\pi/3$

Q9 The general solution of $\sin x - \cos x = \sqrt{2}$ for any integer n is

- (A) $n\pi$
 (B) $2n\pi + \frac{3\pi}{4}$
 (C) $2n\pi$
 (D) $(2n+1)\pi$

Q10 General solution of the equation $\cot \theta - \tan \theta = 2$ is

- (A) $n\pi + \frac{\pi}{4}$
 (B) $\frac{n\pi}{2} + \frac{\pi}{8}$
 (C) $\frac{n\pi}{2} \pm \frac{\pi}{8}$
 (D) None of these

Q11



The number of solution of the equation $\tan x + \sec x = 2 \cos x$ lying in the interval $(0, 2\pi)$ is.

- (A) 0 (B) 1
(C) 2 (D) 3

Q12 The number of solutions of the equation $3 \sin^2 x - 7 \sin x + 2 = 0$ in the interval $[0, 5\pi]$ is

- (A) 0 (B) 5
(C) 6 (D) 10

Q13 The number of solutions of $\sin x + \sin 2x + \sin 3x = \cos x + \cos 2x + \cos 3x$, $0 \leq x \leq 2\pi$, is

- (A) 7 (B) 5
(C) 4 (D) 6

Q14 The number of solutions of the equation $\sin 2\theta - 2 \cos \theta + 4 \sin \theta = 4$ in $[0, 5\pi]$ is equal to

- (A) 3 (B) 4
(C) 5 (D) 6

Q15 The number of solutions of $\cos 2\theta = \sin \theta$ is $(0, 2\pi)$ is —

- (A) 1 (B) 2
(C) 3 (D) No value

Q16 The number of real roots of the equation $\operatorname{cosec} \theta + \sec \theta - \sqrt{15} = 0$ lying in $[0, \pi]$ is

- (A) 6 (B) 8
(C) 4 (D) 0

Q17 Total number of solutions of $\cos x = \sqrt{1 - \sin 2x}$ in $[0, 2\pi]$, is equal to

- (A) 2 (B) 3
(C) 5 (D) None of these

Q18 If $\sin x + \cos x = \sqrt{y + \frac{1}{y}}$, $x \in (0, \pi)$, then

- (A) $x = \frac{\pi}{4}$, $y = 1$

(B) $y = 0$

(C) $y = 2$

(D) $x = \frac{3\pi}{4}$

Q19 If $a, b \in [0, 2\pi]$ and equation $x^2 - 2x + 4 = 3 \sin(ax + b)$ has at least one solution and the least positive value of $a + b$ is k , then $\frac{2k}{\pi}$ is equal

- (A) 0 (B) 1
(C) 2 (D) 3

Q20 Number of ordered pairs (x, y) which satisfies the relation $\frac{x^4+1}{8x^2} = \sin^2 y \cdot \cos^2 y$, where $y \in [0, 2\pi]$

- (A) 4 (B) 6
(C) 8 (D) 10



Answer Key

Q1 (D)

Q2 (D)

Q3 (B)

Q4 (A)

Q5 (B)

Q6 (B)

Q7 (B)

Q8 (D)

Q9 (B)

Q10 (B)

Q11 (C)

Q12 (C)

Q13 (D)

Q14 (A)

Q15 (C)

Q16 (C)

Q17 (B)

Q18 (A)

Q19 (B)

Q20 (C)

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